

Lecture 7: Directions in the Data

In which directions is the cloud of countries spread out, and by how much

STAT 481 — Multivariate Analysis

Tue, Sep 15, 2026 • 50 minutes

Today's Overview

- 1. The data.** The file `data/whwide2022.csv`, 175 countries, four columns: share of the population under 15 (`pop0014`, %), share 65 and over (`pop65`, %), life expectancy (`life_exp`, years) and income per person (`gdp_k`, thousand US\$). Each country is one point with four coordinates; the 175 points together are “the cloud”. We work with their 4×4 covariance table all lecture.
- 2. Make a new column by adding two old ones** (share under 15 + share 65 and over; we call them the *young share* and the *old share*). Its variance can be read straight off the covariance table, without going back to the 175 rows. We check this in R.
- 3. A variance is never negative.** We see what this says about the covariance table, and we find the one case where the answer is exactly zero.
- 4. Find the directions in which the cloud of countries is spread out.** R gives four directions and four numbers (how spread out the cloud is in each direction). This is the main part of the lecture.
- 5. The four numbers add up to the total variance and multiply to the determinant.** Two checks in R.
- 6. Draw the picture.** The countries that are at the same statistical distance from the centre form an ellipse. Its axes are the directions we just found. Niger, Japan and Qatar are placed on it, and the result is surprising.
- 7. Make every column have variance 1 and no correlation.** One matrix does it, and after it the statistical distance is just the ordinary straight-line distance.
- 8. One closing fact:** no way of combining the four columns gives a variance bigger than the first of the four numbers.
- 9. Then:** a warm-up, four home problems on other columns of the same file, and an FAQ.

Learning Objectives

By the end of this lecture you should be able to:

1. write the variance of a weighted total of the variables as $\mathbf{a}^\top \mathbf{S} \mathbf{a}$ and compute it two ways in R;
2. explain why $\mathbf{S}$ is nonnegative definite and when it is positive definite;
3. compute the eigenvalues and eigenvectors of $\mathbf{S}$ in R, fix their signs, and read them as *uncorrelated directions* and *their variances*;
4. use the two identities $\text{trace} = \text{sum of eigenvalues}$, $\text{determinant} = \text{product of eigenvalues}$, and rebuild $\mathbf{S}$ from its eigen-pairs;
5. read the ellipse of constant Mahalanobis distance;
6. build $\mathbf{S}^{-1/2}$ from the eigen-pairs, whiten a data table, and show that the Mahalanobis distance is the straight-line distance after whitening;
7. state which single number bounds the variance of every unit-length combination of the variables.

Packages today

- **Base R** [known]: `eigen()`, `diag()`, `%*%`, `solve()`, `det()`, `mahalanobis()`, `zapsmall()`, `apply()`.
- **car** [NEW]: `ellipse()` draws the ellipse of Section 5 (in `R/L7_figures.R`, not in a printed chunk). `install.packages("car")` once.

Open the RStudio project `multi` first, so that `"data/whwide2022.csv"` resolves. Indicators: World Bank WDI, 2022.

Notation

- $\mathbf{X}$: the 175×4 data table. $\bar{\mathbf{x}}$: the vector of the four column means. $\mathbf{X}_c$: the table with each column mean subtracted ("centred"). $\mathbf{S} = \mathbf{X}_c^\top \mathbf{X}_c / (n - 1)$: the 4×4 covariance matrix. $\mathbf{S}^{-1}$: its inverse. $^\top$: transpose. $\mathbf{I}$: the identity matrix.
- $\mathbf{a} = (a_1, \dots, a_p)^\top$: a vector of **weights**, fixed by us. $\mathbf{x}$: one country's row, written as a column. $\mathbf{a}^\top \mathbf{x} = a_1 x_1 + \dots + a_p x_p$ is one number for that country.
- Two new symbols, λ and $\mathbf{e}$, arrive in Section 3.
- $n = 175$, $p = 4$ throughout, except in the small 2×2 examples.

1 A new column from old ones, and its variance

In this section

The question. *If I add two columns — the share under 15 plus the share over 65 — I get a new column. Do I need the 175 rows to find its variance, or is the answer already in the covariance table?*

What we do about it. Write the new column as a weighted total, give the one-line formula for its variance, and check it in R on a sum and on a difference.

1. Definition

Definition 1.1: Weighted total (linear combination) and its variance

- Pick p weights $a_1, \dots, a_p$ and put them in a vector $\mathbf{a}$. For one country with values $x_1, \dots, x_p$, the **weighted total** is one number:

$$y = \mathbf{a}^\top \mathbf{x} = a_1 x_1 + a_2 x_2 + \dots + a_p x_p.$$

Do this for every country and you get a new column of n numbers.

- **Its variance can be computed from the covariance matrix alone:**

$$\text{Var}(y) = \mathbf{a}^\top \mathbf{S} \mathbf{a}.$$

There is no need to go back to the rows.

- **Where the formula comes from.** y is a sum of the x_j with weights a_j , so $\text{Var}(y) = \sum_j a_j^2 s_{jj} + 2 \sum_{j < k} a_j a_k s_{jk}$. Writing this with matrices gives $\mathbf{a}^\top \mathbf{S} \mathbf{a}$.
- **Small example.** Two variables, each with variance 1, covariance 0.5. The weights of their *sum* are $\mathbf{a} = (1, 1)^\top$:

$$\text{Var}(x_1 + x_2) = \mathbf{a}^\top \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix} \mathbf{a} = 1 + 1 + 2(0.5) = 3.$$

- A number of the form $\mathbf{a}^\top \mathbf{A} \mathbf{a}$ (a symmetric matrix with a vector on each side) is called a **quadratic form**: it only contains squares a_j^2 and products $a_j a_k$.

2. In R

Read the file and build the four columns. Income is divided by 1 000 so it is in thousand dollars.

```
options(digits = 4)
w <- read.csv("data/whwide2022.csv", stringsAsFactors = FALSE)
X <- as.matrix(w[, c("pop0014", "pop65", "life_exp", "gdp_pc")])
X[, "gdp_pc"] <- X[, "gdp_pc"] / 1000 # income in THOUSAND US$
colnames(X)[4] <- "gdp_k"
rownames(X) <- w$country
n <- nrow(X); p <- ncol(X)
Xc <- scale(X, center = TRUE, scale = FALSE) # each column minus its mean
S <- cov(X) # the covariance matrix
```

- `scale(X, center = TRUE, scale = FALSE)` subtracts each column's mean and nothing else. `cov(X)` is the covariance matrix **S**; here it is:

```
round(S, 2)
```

	pop0014	pop65	life_exp	gdp_k
pop0014	110.22	-61.31	-70.56	-160.7
pop65	-61.31	51.03	38.68	119.3
life_exp	-70.56	38.68	67.28	130.2
gdp_k	-160.73	119.28	130.24	801.6

- The diagonal holds the four variances: income (in thousands) 801.6, the two age shares 110.2 and 51.0, life expectancy 67.3.
- The entry -61.31 says the young share and the old share move in opposite directions: a country with many children has few elderly.

The small example of the definition, typed exactly as in the definition:

```
Sa <- matrix(c(1, 0.5, 0.5, 1), 2) # variances 1 and 1, covariance 0.5
a <- c(1, 1) # weights of the sum
```

```
t(a) %*% Sa %*% a
```

```
      [,1]
[1,]      3
```

- `t(a) %*% Sa %*% a` is $(1 \times 2)(2 \times 2)(2 \times 1)$, so a 1×1 matrix. It prints 3, as on paper.

? Guess first 1 — answer before you read on

We build two new columns from the real data: the *sum* young share + old share, and the *difference* young share – old share. Both use the variances 110.22 and 51.03 and the covariance -61.31 . Which of the two new columns has the bigger variance? Write down your guess.

The sum, two ways — from the 175 numbers, then from the table:

```
a <- c(1, 1, 0, 0) # weights: young share + old share
y <- X %*% a # the new column: one number per country
```

```
var(y)
```

```
      [,1]  
[1,] 38.63
```

```
t(a) %*% S %*% a
```

```
      [,1]  
[1,] 38.63
```

- $X \%*\% a$ multiplies every row by the weights: it is the new column of 175 numbers. `var()` of that column is 38.63.
- $t(a) \%*\% S \%*\% a$ never looks at the rows and gives the same 38.63: $110.22 + 51.03 - 2(61.31) = 38.63$.

Now the difference:

```
b <- c(1, -1, 0, 0)           # weights: young share - old share
```

```
var(X %*% b)
```

```
      [,1]  
[1,] 283.9
```

```
t(b) %*% S %*% b
```

```
      [,1]  
[1,] 283.9
```

- Again the two ways agree: $110.22 + 51.03 + 2(61.31) = 283.9$.
- **The difference is seven times more variable than the sum.** Young and old shares move in opposite directions, so when one goes up the other goes down: their sum moves very little, their difference moves twice as much.

2 That variance is never negative

In this section

The question. A variance cannot be negative. So $\mathbf{a}^\top \mathbf{S} \mathbf{a} \geq 0$ for every choice of weights. Can it ever be exactly zero?

What we do about it. Give the two names (nonnegative definite, positive definite), say what they mean for data, and build a table in R whose quadratic form is zero.

1. Definition

Definition 2.1: Nonnegative definite; positive definite

- A symmetric matrix $\mathbf{A}$ is **nonnegative definite** if its quadratic form is never negative:

$$\mathbf{a}^\top \mathbf{A} \mathbf{a} \geq 0 \quad \text{for every } \mathbf{a}.$$

- It is **positive definite** if the only way to get 0 is to take all weights equal to 0:

$$\mathbf{a}^\top \mathbf{A} \mathbf{a} > 0 \quad \text{for every } \mathbf{a} \neq \mathbf{0}.$$

Definition 2.1: Nonnegative definite; positive definite (continued)

- **Symmetric alone is not enough.** $\mathbf{B} = \begin{pmatrix} 1 & -5 \\ -5 & 1 \end{pmatrix}$ is symmetric, but with $\mathbf{a} = (1, 1)^\top$ its quadratic form is $1 + 1 - 10 = -8$. So $\mathbf{B}$ can never be a covariance matrix.

2. In statistics

Fact 2.1: Every covariance matrix is nonnegative definite

- $\mathbf{a}^\top \mathbf{S} \mathbf{a}$ is the variance of a column, and a variance is a sum of squares divided by $n - 1$. So it is ≥ 0 for every $\mathbf{a}$, for any data set.
- $\mathbf{a}^\top \mathbf{S} \mathbf{a} = 0$ means the new column has variance 0: it is **the same number for every country**. That happens only when one column of the table can be computed exactly from the others.
- So: $\mathbf{S}$ is **positive definite** exactly when no column is a combination of the others. Then $|\mathbf{S}| > 0$ and $\mathbf{S}^{-1}$ exists. If one column is a combination of the others, $|\mathbf{S}| = 0$ and there is no inverse.

3. In R

? Guess first 2 — answer before you read on

Add a fifth column `dep = young share + old share` to the table, and take the weights $\mathbf{a}_0 = (1, 1, 0, 0, -1)$. The new column is `young + old - dep`. What is its value for Qatar? For Niger? What will `var()` print?

```
X5 <- cbind(X, dep = X[, "pop0014"] + X[, "pop65"]) # fifth column = sum
S5 <- cov(X5)
a0 <- c(1, 1, 0, 0, -1) # young + old - dep
```

```
var(X5 %*% a0)
```

```
      [,1]
[1,]      0
```

```
t(a0) %*% S5 %*% a0
```

```
      [,1]
[1,] 8.171e-14
```

- The new column is 0 for every country, so its variance is 0.
- The quadratic form prints 8×10^{-14} : this is 0 plus a tiny rounding error from the computer. **This five-column table is nonnegative definite but not positive definite.** It has no inverse.

3 Eigenvalues and eigenvectors: the directions of the cloud

In this section

The question. *The four columns are correlated. Is there a way to turn the table into four NEW columns that are not correlated at all — and what are their variances?*

What we do about it. Define an eigenvalue and an eigenvector on a 2×2 matrix by hand, then let `eigen()` do the 4×4 case. The eigenvectors are the uncorrelated directions and the eigenvalues are their variances. **What you keep from this section:** the objects that every later section uses.

1. Definition

Definition 3.1: Eigenvalue, eigenvector

- Take a square matrix $\mathbf{A}$. A number λ and a non-zero vector $\mathbf{e}$ form an **eigenvalue–eigenvector pair** if

$$\mathbf{A}\mathbf{e} = \lambda \mathbf{e}.$$

In words: multiplying $\mathbf{e}$ by $\mathbf{A}$ does not change its direction; it only multiplies its length by λ .

- We always make $\mathbf{e}$ have **length 1**: $\mathbf{e}^\top \mathbf{e} = e_1^2 + \cdots + e_p^2 = 1$.
- The sign is free.** If $\mathbf{e}$ works, so does $-\mathbf{e}$. Two programs may print opposite signs for the same eigenvector; neither is wrong.
- A symmetric $p \times p$ matrix has p eigenvalues, all real, and its eigenvectors are **perpendicular** to each other: $\mathbf{e}_i^\top \mathbf{e}_j = 0$ for $i \neq j$.
- Small example by hand.** $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Try $\mathbf{e} = (1, 1)^\top$:

$$\mathbf{M} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1 \\ 1+2 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

So $\lambda_1 = 3$ with $\mathbf{e}_1 = (1, 1)^\top / \sqrt{2} = (0.707, 0.707)^\top$. Try $\mathbf{e} = (1, -1)^\top$:

$$\mathbf{M} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2-1 \\ 1-2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

So $\lambda_2 = 1$ with $\mathbf{e}_2 = (1, -1)^\top / \sqrt{2}$. The two vectors are perpendicular: $1 \cdot 1 + 1 \cdot (-1) = 0$.

- How to find them without guessing.** The eigenvalues are the solutions of $|\mathbf{A} - \lambda \mathbf{I}| = 0$. For $\mathbf{M}$: $(2 - \lambda)^2 - 1 = 0$, so $\lambda = 3$ or 1 . For 4×4 we let R do it.

2. In statistics

Fact 3.1: Eigenvectors are the uncorrelated directions; eigenvalues are their variances

- An eigenvector $\mathbf{e}_i$ of $\mathbf{S}$ is a list of p weights — a **direction**. Use it as weights: $z_i = \mathbf{e}_i^\top \mathbf{x}$ is a new column, one number per country.
- **Its variance is the eigenvalue.** By Section 1 with $\mathbf{a} = \mathbf{e}_i$:

$$\text{Var}(z_i) = \mathbf{e}_i^\top \mathbf{S} \mathbf{e}_i = \mathbf{e}_i^\top (\lambda_i \mathbf{e}_i) = \lambda_i (\mathbf{e}_i^\top \mathbf{e}_i) = \lambda_i.$$

- **Two different eigenvectors give uncorrelated columns:**

$$\text{Cov}(z_i, z_j) = \mathbf{e}_i^\top \mathbf{S} \mathbf{e}_j = \lambda_j \mathbf{e}_i^\top \mathbf{e}_j = 0 \quad (i \neq j).$$

- Put the eigenvectors side by side as the columns of a matrix $\mathbf{E} = (\mathbf{e}_1, \dots, \mathbf{e}_p)$. Then $\mathbf{Z} = \mathbf{X}_c \mathbf{E}$ is the table of the p new columns, and its covariance matrix is **diagonal**, with $\lambda_1, \dots, \lambda_p$ on the diagonal.
- Because each λ_i is a variance, $\lambda_i \geq 0$ for a covariance matrix; $\lambda_i > 0$ for all i when $\mathbf{S}$ is positive definite.

3. In R

The 2×2 example first. `eigen(A)` returns a list with two parts: `$values`, the eigenvalues in **decreasing** order, and `$vectors`, a matrix whose **columns** are the eigenvectors, each of length 1.

```
M <- matrix(c(2, 1, 1, 2), 2)
```

```
eigen(M)
```

```
eigen() decomposition
$values
[1] 3 1

$vectors
      [,1] [,2]
[1,] 0.7071 -0.7071
[2,] 0.7071  0.7071
```

- Values 3 and 1, as by hand. The first column is $(0.707, 0.707)^\top = (1, 1)^\top / \sqrt{2}$; the second is $(-0.707, 0.707)^\top$, which is $-\mathbf{e}_2$: same eigenvector, opposite sign.

One line to check the definition $\mathbf{M}\mathbf{e}_1 = 3\mathbf{e}_1$:

```
M %*% c(1, 1) / sqrt(2)
```

```
      [,1]
[1,] 2.121
[2,] 2.121
```

- $2.121 = 3 \times 0.707$ in both entries: the vector is multiplied by 3 and keeps its direction.

? Guess first 3 — answer before you read on

Now `eigen(S)` on the four real columns. The diagonal of **S** is 110, 51, 67, 802. Which variable do you expect to have the biggest weight in the first eigenvector? Roughly how big will the first eigenvalue be?

```
ev      <- eigen(S)
lambda  <- ev$values
E       <- ev$vectors
```

```
lambda
```

```
[1] 884.575 115.943 20.078 9.571
```

- Four eigenvalues, from largest to smallest: 884.6, 115.9, 20.1, 9.6. The first is bigger than any variance on the diagonal.

```
round(E, 3)
```

	[,1]	[,2]	[,3]	[,4]
[1,]	0.225	0.726	0.160	0.629
[2,]	-0.160	-0.372	-0.643	0.650
[3,]	-0.178	-0.477	0.749	0.424
[4,]	-0.945	0.326	0.006	-0.040

- Four columns, one eigenvector each. The first column is $(0.225, -0.160, -0.178, -0.945)^\top$: almost all of its weight is on the fourth row, income.
- R chose the sign that makes income negative. We flip the sign of any column whose biggest entry is negative, so that every printout looks the same. Nothing else changes.

```
for (j in 1:p) # biggest entry of column j negative? flip it
  if (E[which.max(abs(E[, j])), j] < 0) E[, j] <- -E[, j]
dimnames(E) <- list(colnames(X), paste0("e", 1:p))
```

```
round(E, 3)
```

	e1	e2	e3	e4
pop0014	-0.225	0.726	0.160	0.629
pop65	0.160	-0.372	-0.643	0.650
life_exp	0.178	-0.477	0.749	0.424
gdp_k	0.945	0.326	0.006	-0.040

- `which.max(abs(E[, j]))` finds the row of the biggest entry of column j ; if that entry is negative the whole column is multiplied by -1 . `dimnames` puts the variable names on the rows and **e1**, ..., **e4** on the columns.
- **e1** = $(-0.22, 0.16, 0.18, \mathbf{0.94})^\top$ with $\lambda_1 = 884.6$: the first direction is **almost only income**. Income has a much bigger variance than the other three columns (it is in thousands of dollars, the others are in percent or years), so the cloud is spread out mostly along income.
- **e2** = $(0.73, -0.37, -0.48, 0.33)^\top$ with $\lambda_2 = 115.9$: **young** countries against **old, long-lived** countries — the age pattern, after income is removed. **e3** and **e4** have only 20 and 9.6: very little variance is left.

Are the four eigenvectors really perpendicular and of length 1?

```
zapsmall(t(E) %*% E)
```

```
      e1 e2 e3 e4
e1    1  0  0  0
e2    0  1  0  0
e3    0  0  1  0
e4    0  0  0  1
```

- $\mathbf{E}^\top \mathbf{E}$ is the identity: every column has length 1 (the diagonal) and every pair is perpendicular (the zeros). `zapsmall()` prints tiny rounding errors as 0.

? Guess first 4 — answer before you read on

Build the four new columns $\mathbf{Z} = \mathbf{X}_c \mathbf{E}$ and ask for `cov(Z)`. What do you expect on the diagonal? Off the diagonal?

```
Z <- Xc %*% E # four new columns, one per eigenvector
colnames(Z) <- paste0("z", 1:p)
```

```
var(Z[, 1])
```

```
[1] 884.6
```

- `Xc %*% E` multiplies every centred row by all four eigenvectors in one step: one new column per eigenvector. The variance of the first new column is $884.6 = \lambda_1$, exactly.

```
zapsmall(cov(Z))
```

```
      z1      z2      z3      z4
z1 884.6    0.0    0.0    0.0
z2   0.0  115.9    0.0    0.0
z3   0.0    0.0   20.1    0.0
z4   0.0    0.0    0.0    9.6
```

- **Diagonal.** The four eigenvalues are on the diagonal and every covariance is 0: the new columns are uncorrelated. This is the answer to the question of the section.

```
round(Z[c("Qatar", "Japan", "Niger"), ], 2)
```

```
      z1      z2      z3      z4
Qatar 69.23  13.29  10.98 -11.42
Japan 24.65 -17.93  -6.30   7.68
Niger -24.81  17.71  -1.25   3.94
```

- Qatar's $z_1 = 69$ is far along the income direction (its income is about 70 thousand above the average). Niger and Japan are at +18 and -18 on z_2 , the age pattern: opposite sides.

4 The eigenvalues add to the total variance and multiply to the determinant

In this section

The question. *Do the four eigen-pairs know everything about $\mathbf{S}$?*

What we do about it. Yes: $\mathbf{S}$ can be rebuilt from them. Two consequences are checked in R — trace = sum of eigenvalues, determinant = product — and a determinant of 0 is explained.

1. Definition

Definition 4.1: Spectral decomposition

- Put the eigenvalues in a diagonal matrix $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p)$ and the eigenvectors as the columns of $\mathbf{E}$. Then

$$\mathbf{S} = \mathbf{E} \mathbf{\Lambda} \mathbf{E}^\top = \lambda_1 \mathbf{e}_1 \mathbf{e}_1^\top + \lambda_2 \mathbf{e}_2 \mathbf{e}_2^\top + \dots + \lambda_p \mathbf{e}_p \mathbf{e}_p^\top.$$

The matrix is a sum of p pieces, one per direction, each weighted by its eigenvalue.

- Trace = sum of eigenvalues.**

$$\text{tr}(\mathbf{S}) = s_{11} + s_{22} + \dots + s_{pp} = \lambda_1 + \lambda_2 + \dots + \lambda_p.$$

- Determinant = product of eigenvalues.**

$$|\mathbf{S}| = \lambda_1 \lambda_2 \dots \lambda_p.$$

- Small example.** For $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with $\lambda = 3, 1$: trace $2 + 2 = 4 = 3 + 1$; determinant $4 - 1 = 3 = 3 \times 1$.

2. In statistics

Fact 4.1: Total variance is the sum; one zero eigenvalue makes the determinant zero

- The **total variance** $\text{tr}(\mathbf{S})$ is also the sum of the variances of the p uncorrelated new columns. So $\lambda_i / \sum \lambda$ is the **share of the total variance** that lies in direction i .
- $|\mathbf{S}| = \prod \lambda_i$ is 0 as soon as *one* eigenvalue is 0: one direction with zero variance is enough. That is why one column that repeats the others (Section 2) makes the determinant 0, and then there is no inverse.

3. In R

```
c(sum_lambda = sum(lambda), trace_S = sum(diag(S)))
```

```
sum_lambda    trace_S
      1030         1030
```

- `sum(diag(S))` adds the four variances on the diagonal: $110.2 + 51.0 + 67.3 + 801.6 = 1030$. The four eigenvalues add to the same 1030.

```
c(prod_lambda = prod(lambda), det_S = det(S))
```

```
prod_lambda    det_S
 19709199      19709199
```

- The product of the four eigenvalues equals `det(S)`: 19 709 199 both ways.

```
round(lambda / sum(lambda), 3)
```

```
[1] 0.859 0.113 0.019 0.009
```

- 86% of the total variance lies along $\mathbf{e}_1$ (income), 11% along $\mathbf{e}_2$ (the age contrast), under 3% in the other two directions together.

```
all.equal(E %*% diag(lambda) %*% t(E), S, check.attributes = FALSE)
```

```
[1] TRUE
```

- `diag(lambda)` builds $\mathbf{\Lambda}$; `E %*% diag(lambda) %*% t(E)` is $\mathbf{E}\mathbf{\Lambda}\mathbf{E}^\top$, and it equals $\mathbf{S}$. **The four eigenpairs are $\mathbf{S}$** , written another way.

? Guess first 5 — answer before you read on

Back to the five-column table $\mathbf{S5}$ of Section 2, which has one column that is the sum of two others. What will `eigen(S5)$values` print in last place? Will `det(S5)` be exactly 0?

```
eigen(S5)$values
```

```
[1] 8.885e+02 1.333e+02 3.079e+01 1.621e+01 8.715e-14
```

- Four real variances and then 8.7×10^{-14} : this is a zero, printed with rounding error. One direction with zero variance.

```
c(prod_lambda = prod(eigen(S5)$values), det_S5 = det(S5))
```

```
prod_lambda      det_S5
5.153e-06        1.890e-06
```

- The product of the five eigenvalues is 5.2×10^{-6} ; `det()` says 1.9×10^{-6} . **They do not agree, and neither number means anything:** both are (a rounding error) $\times$ (four real eigenvalues), computed in two different ways. The true determinant of this table is 0.

5 The countries at one statistical distance form an ellipse

In this section

The question. Take all the points that are at the same Mahalanobis distance from the centre of the cloud. What shape do they make?

What we do about it. Show that the shape is an ellipse whose axes are the eigenvectors and whose half-lengths are $\sqrt{\lambda_i}$; draw it on two of the four columns; and put Niger, Japan and Qatar on the picture with their straight-line and their Mahalanobis distances side by side.

1. Definition

Definition 5.1: The ellipse of constant Mahalanobis distance

- The squared Mahalanobis distance of a point $\mathbf{x}$ from the centre $\bar{\mathbf{x}}$ is $(\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}})$. The points at squared distance c^2 form the set

$$\{\mathbf{x} : (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = c^2\}.$$

- It is an ellipse.** Measure the centred point along the eigenvectors: $y_i = \mathbf{e}_i^\top (\mathbf{x} - \bar{\mathbf{x}})$. Because $\mathbf{S}^{-1} = \mathbf{E}\mathbf{\Lambda}^{-1}\mathbf{E}^\top$ (apply $1/\lambda_i$ to each eigenvalue), the equation becomes

$$\frac{y_1^2}{\lambda_1} + \frac{y_2^2}{\lambda_2} + \dots + \frac{y_p^2}{\lambda_p} = c^2.$$

For $p = 2$ this is the equation of an ellipse with **axes along** $\mathbf{e}_1$, $\mathbf{e}_2$ and **half-lengths** $c\sqrt{\lambda_1}$, $c\sqrt{\lambda_2}$.

- Small example.** $\lambda = 4, 1$ and $c = 1$: half-lengths 2 and 1; the long axis is along $\mathbf{e}_1$.

2. In statistics

Fact 5.1: Along the long axis, a big step is a small Mahalanobis distance

- Along $\mathbf{e}_1$ the data vary a lot (λ_1 large). A country can be far from the centre in ordinary units and still be at a small Mahalanobis distance.
- Along $\mathbf{e}_p$ the data vary very little. A small step in ordinary units is a large Mahalanobis distance.
- Mahalanobis distance is measured in units of spread, direction by direction.** Two countries at the same straight-line distance from the centre can be at very different Mahalanobis distances.

3. In R

The picture needs two columns. We take the two age shares — both in percentage points, so the two axes have the same unit.

```
P2 <- Xc[, c("pop0014", "pop65")] # two centred columns, both in %
S2 <- cov(P2)
```

```
round(S2, 2)
```

```
      pop0014  pop65
pop0014 110.22 -61.31
pop65   -61.31  51.03
```

- The 2×2 covariance matrix of the pair: the top-left block of $\mathbf{S}$.

```
ev2 <- eigen(S2)
lam2 <- ev2$values
E2 <- ev2$vectors
for (j in 1:2)
  if (E2[which.max(abs(E2[, j])), j] < 0) E2[, j] <- -E2[, j]
```

```
lam2
```

```
[1] 148.71 12.55
```

- Eigenvalues 148.7 and 12.6: the cloud is about twelve times more stretched in one direction than in the other.

```
round(E2, 3)
```

```
      [,1] [,2]
[1,]  0.847 0.532
[2,] -0.532 0.847
```

- $\mathbf{e}_1 = (0.85, -0.53)^\top$ is “young minus old”; $\mathbf{e}_2 = (0.53, 0.85)^\top$ is “young plus old”. Along the second the variance is only 12.6: **the total share of children plus elderly varies little from country to country**; what varies is whether they are children or elderly.

```
sqrt(lam2)
```

```
[1] 12.195  3.542
```

- Half-lengths of the $c = 1$ ellipse: $\sqrt{148.7} = 12.2$ and $\sqrt{12.6} = 3.5$ percentage points.

? Guess first 6 — answer before you read on

Niger has many children and few elderly. Qatar has few children *and* few elderly (its population is mostly adult migrant workers). Which of the two is farther from the centre in a straight line? And in Mahalanobis distance? Write both guesses down.

```
three <- c("Niger", "Japan", "Qatar")
dE <- sqrt(rowSums(P2^2))                # straight line from centre
dM <- sqrt(mahalanobis(P2, center = c(0, 0), cov = S2)) # Mahalanobis
```

```
round(cbind(P2, d_E = dE, d_M = dM)[three, ], 2)
```

	pop0014	pop65	d_E	d_M
Niger	20.73	-7.19	21.94	2.24
Japan	-14.74	19.66	24.57	3.12
Qatar	-11.35	-8.35	14.09	3.72

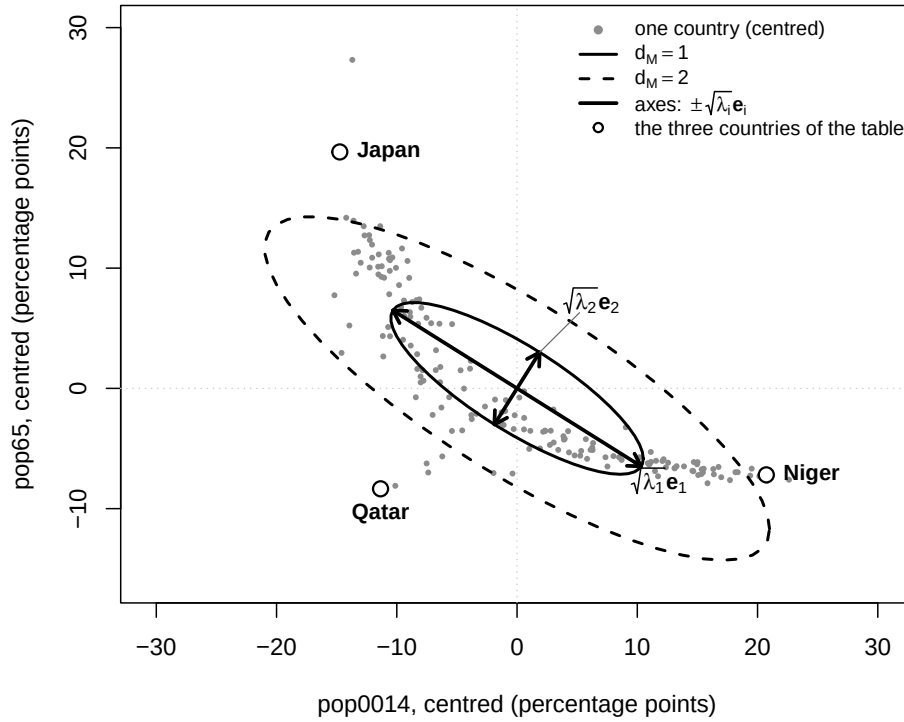
- The squared straight-line distance of each centred row from the origin is `rowSums(P2^2)`; the squared Mahalanobis distance is `mahalanobis(x, center, cov)`. `sqrt()` of both gives the distances.
- **Niger is farther in a straight line (21.9 against 14.1) and nearer in Mahalanobis distance (2.2 against 3.7).** Niger’s step (+20.7 young, −7.2 old) is along $\mathbf{e}_1$, the long axis, where the data vary a lot. Qatar is below the centre on *both* shares: a step along $\mathbf{e}_2$, where one percentage point counts more than three times as much in Mahalanobis distance.
- Japan is the farthest of the three in a straight line (24.6), and Qatar has the larger Mahalanobis distance (3.7 against 3.1).

How to read this figure

- **One grey dot** is one country: share under 15 minus 26.6 (horizontal), share over 65 minus 9.7 (vertical). Both axes are percentage points, drawn with equal scales, so the shape is not distorted.
- **The solid ellipse** is every point at Mahalanobis distance 1 from the centre; **the dashed one** at distance 2 (the solid one, doubled).

How to read this figure (continued)

- **The two double arrows** are the axes of the solid ellipse: the long one (12.2 units) runs down-right along “young minus old”; the short one (3.5 units) runs up-right along “young plus old”. They are perpendicular.
- The dotted lines through 0 only mark the centre.
- **What to look for, in order:** (1) the points form a long, tilted shape, not a circle; (2) the arrows point along that shape; (3) Niger is far from the centre but *along* the shape, Qatar is *outside* the shape — count how many ellipses each one has crossed.
- Drawn by R/L7_figures.R with `car::ellipse(center, shape, radius)`: the centre, the covariance matrix, and c .



6 Whitening: the transformation behind the Mahalanobis distance

In this section

The question. *Is there one matrix that turns the four correlated columns into four columns with variance 1 and no correlation at all?*

What we do about it. Build $\mathbf{S}^{-1/2}$ from the eigen-pairs, apply it to the centred table, and watch the covariance matrix become the identity. After that, the Mahalanobis distance is just the straight-line distance.

1. Definition

Definition 6.1: Inverse and inverse square root of $\mathbf{S}$

- Keep the eigenvectors, change only the eigenvalues:

$$\mathbf{S}^{-1} = \mathbf{E} \operatorname{diag}\left(\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_p}\right) \mathbf{E}^\top, \quad \mathbf{S}^{-1/2} = \mathbf{E} \operatorname{diag}\left(\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_p}}\right) \mathbf{E}^\top.$$

- $\mathbf{S}^{-1/2}$ is symmetric, and $\mathbf{S}^{-1/2} \mathbf{S}^{-1/2} = \mathbf{S}^{-1}$ (multiply: the middle $\mathbf{E}^\top \mathbf{E}$ is $\mathbf{I}$, and $\frac{1}{\sqrt{\lambda_i}} \cdot \frac{1}{\sqrt{\lambda_i}} = \frac{1}{\lambda_i}$).

Definition 6.1: Inverse and inverse square root of $\mathbf{S}$ (continued)

- **Small example.** $\mathbf{M}$ with $\lambda = 3, 1$: $\mathbf{M}^{-1/2}$ has eigenvalues $1/\sqrt{3}$ and 1, same eigenvectors.

2. In statistics

Fact 6.1: Whitening: Mahalanobis distance is the straight-line distance after $\mathbf{S}^{-1/2}$

- Multiply every centred row by $\mathbf{S}^{-1/2}$; all rows in one step: $\mathbf{Z}_w = \mathbf{X}_c \mathbf{S}^{-1/2}$.
- The new table has covariance matrix

$$\text{Cov}(\mathbf{Z}_w) = \mathbf{S}^{-1/2} \mathbf{S} \mathbf{S}^{-1/2} = \mathbf{I} :$$

every new column has variance 1 and no two are correlated. This is called **whitening**.

- The squared length of a whitened row *is* the squared Mahalanobis distance:

$$\mathbf{z}^\top \mathbf{z} = (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1/2} \mathbf{S}^{-1/2} (\mathbf{x} - \bar{\mathbf{x}}) = (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x} - \bar{\mathbf{x}}).$$

3. In R

```
S_ihalf <- E %%% diag(1 / sqrt(lambda)) %%% t(E) # S^{-1/2}
```

```
all.equal(S_ihalf %%% S_ihalf, solve(S), check.attributes = FALSE)
```

```
[1] TRUE
```

- `diag(1 / sqrt(lambda))` holds $1/\sqrt{\lambda_i}$; the sandwich `E %%% ... %%% t(E)` is the definition. Squared, it equals `solve(S)`, the inverse.

? Guess first 7 — answer before you read on

Whiten the 175 centred rows: `Zw <- Xc %%% S_ihalf`. What will `cov(Zw)` look like?

```
Zw <- Xc %%% S_ihalf # every country, whitened
```

```
zapsmall(cov(Zw))
```

```
      pop0014 pop65 life_exp gdp_k
pop0014      1    0      0      0
pop65        0    1      0      0
life_exp     0    0      1      0
gdp_k        0    0      0      1
```

- The 4×4 identity: four columns of variance 1, no correlation. Whitened.

```
d2_w <- rowSums(Zw^2) # squared length of each whitened row
```

```
all.equal(d2_w, mahalanobis(X, center = colMeans(X), cov = S))
```

```
[1] TRUE
```

- The squared straight-line length of each whitened row equals `mahalanobis()` for all 175 countries: whitening gives the Mahalanobis distance.

```
d2_w["Qatar"]
```

```
Qatar
26.57
```

- Qatar's squared Mahalanobis distance from the centre, on the four columns, is 26.57.

7 Closing fact: no combination has a variance above λ_1

In this section

The question. Among all weighted totals with weights of length 1, which one has the biggest variance?

What we do about it. State the answer, then try 5 000 random directions and see that none of them gives a larger variance. We use this fact; we do not prove it.

1. Statement

Fact 7.1: Largest and smallest variance over all unit-length weights

- With $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$,

$$\max_{\mathbf{a}^\top \mathbf{a} = 1} \mathbf{a}^\top \mathbf{S} \mathbf{a} = \lambda_1 \text{ at } \mathbf{a} = \mathbf{e}_1, \quad \min_{\mathbf{a}^\top \mathbf{a} = 1} \mathbf{a}^\top \mathbf{S} \mathbf{a} = \lambda_p \text{ at } \mathbf{a} = \mathbf{e}_p.$$

- **The idea in one line.** $\mathbf{a}^\top \mathbf{S} \mathbf{a}$ is a weighted average of the eigenvalues (the weights are $(\mathbf{e}_i^\top \mathbf{a})^2$ and add up to 1), and an average can never be larger than the largest value in it.
- This fact is the basis of the second half of the course. That is all we say about it today.

2. In R

? Guess first 8 — answer before you read on

Take 5 000 random directions of length 1 and compute $\mathbf{a}^\top \mathbf{S} \mathbf{a}$ for each. Will the largest of the 5 000 be above or below $\lambda_1 = 884.6$? Will the smallest be above or below $\lambda_4 = 9.571$?

```
set.seed(481)
A <- matrix(rnorm(5000 * p), nrow = 5000)      # 5000 random directions
A <- A / sqrt(rowSums(A^2))                     # every row: length 1
q <- apply(A, 1, function(a) t(a) %*% S %*% a) # 5000 variances
```

```
c(smallest = min(q), largest = max(q))
```

```
smallest  largest
   9.67    880.18
```

- `set.seed(481)` fixes the random numbers; `rnorm(5000 * p)` draws 20 000 normal values into a 5000×4 matrix; dividing each row by its length makes it a unit vector; `apply(A, 1, f)` runs `f` on every row.

- Smallest 9.67, largest 880.18.

```
c(lambda_4 = lambda[4], lambda_1 = lambda[1])
```

```
lambda_4 lambda_1
  9.571   884.575
```

- $880.18 < 884.575$ and $9.67 > 9.571$: five thousand random directions came close to both limits and did not pass either of them.

```
t(E[, 1]) %*% S %*% E[, 1]
```

```
      [,1]
[1,] 884.6
```

- At $\mathbf{a} = \mathbf{e}_1$ the variance is exactly λ_1 . The maximum is reached at the eigenvector itself.

8 Warm-up

In this section

The question. *What do you remember?*

What we do about it. Five lines of R on the 2×2 matrix $\mathbf{N} = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$. Guess each output, then run it. The answers follow — try the five first.

Warm-up — five lines, guess each output before you press Enter

1. `N <- matrix(c(5, 2, 2, 2), 2); eigen(N)$values`
2. `sum(diag(N))`
3. `det(N)`
4. `eigen(diag(c(5, 2)))$values`
5. `e <- eigen(N)$vectors[, 1]; t(e) %*% N %*% e`

```
N <- matrix(c(5, 2, 2, 2), 2); eigen(N)$values
```

```
[1] 6 1
```

```
sum(diag(N))
```

```
[1] 7
```

```
det(N)
```

```
[1] 6
```

```
eigen(diag(c(5, 2)))$values
```

```
[1] 5 2
```

```
e <- eigen(N)$vectors[, 1]; t(e) %*% N %*% e
```

```
      [,1]  
[1,]      6
```

- $(5 - \lambda)(2 - \lambda) - 4 = \lambda^2 - 7\lambda + 6 = (\lambda - 6)(\lambda - 1)$: eigenvalues 6 and 1. They add to the trace 7 and multiply to the determinant 6.
- A diagonal matrix has its diagonal as eigenvalues.
- The quadratic form at $\mathbf{e}_1$ is $\lambda_1 = 6$.

9 Problems to try at home — all in R

In this section

The question. *Does everything from today work on other columns of the same file?*

What we do about it. Four problems on three columns of `whwide2022` not used above, plus the correlation matrix of today's four. Each asks you to run something and write **two sentences** about the output. Short answers at the end of the section.

The columns: `birth_rate` and `death_rate` (per 1000 people), `internet_pct` (%):

```
U <- as.matrix(w[, c("birth_rate", "death_rate", "internet_pct")])  
rownames(U) <- w$country
```

Home Problem 1. Directions and their variances. Run `eigen(cov(U))`. Print the eigenvalues and the (rounded) eigenvectors; check that the eigenvalues add to `sum(diag(cov(U)))` and multiply to `det(cov(U))`; compute each eigenvalue's share of the total; build `Z` from the centred `U` and the eigenvectors and print `zapsmall(cov(Z))`. Two sentences: *which column dominates the first direction, and what share of the total variance does that direction carry?*

Home Problem 2. Correlation instead of covariance. Repeat the eigen-analysis on `cor(U)` instead of `cov(U)`, and then on `cor(X)` for the lecture's four columns. Two sentences: *do the eigenvectors change — and what is the first direction of `cor(X)` made of, compared with the $\mathbf{e}_1$ of Section 3?*

Home Problem 3. Positive definite or not. (a) Build `V2` from `urban_pct` and `rural_pct` and print `eigen(cov(V2))$values`. (b) Draw 5000 random unit vectors of length 3 as in Section 7 and compute $\mathbf{a}^T \mathbf{S} \mathbf{a}$ on `cov(U)`; compare `min()` and `max()` with the smallest and largest eigenvalues of Problem 1. Two sentences: *what does the second eigenvalue of (a) look like, and how close did the random vectors get in (b)?*

Home Problem 4. Whiten `U`. Build $\mathbf{S}^{-1/2}$ for `cov(U)` from the eigen-pairs, whiten the centred `U`, print `zapsmall(cov())` of the result, and check `rowSums(Zw^2)` against `mahalanobis(U, colMeans(U), cov(U))` with `all.equal()`. Report Qatar's value. Two sentences: *what is the covariance matrix after whitening, and what does Qatar's number measure?*

Short answers. 1) Eigenvalues 724.9, 25.9, 19.6, summing to 770.4 and multiplying to 367871; the first eigenvector is (0.33, 0.005, -0.94) — `internet_pct` with a little `birth_rate` — and carries 94% of the total; `cov(Z)` is diagonal. Units again: the internet share ranges over 93 points, the two rates over far less. 2) Yes, completely. On `cor(U)` the first direction is (0.71, 0.00, -0.71) — birth rate against internet, equal weights — with share 62%, and `death_rate` gets a direction of its own. On `cor(X)` the first direction is (0.53, -0.51, -0.51, -0.44): all four columns with nearly equal weight, “young versus everything else”, share 75% — the income direction of Section 3 was caused by the units. 3) (a) 974.9 and -6×10^{-14} : a zero, and rounding even gave it the wrong sign — `urban + rural = 100` is one direction with zero variance. (b) `min` 19.62 against $\lambda_3 = 19.59$, `max` 724.73 against $\lambda_1 = 724.94$ — inside the bounds, close to both. 4) The 3×3 identity; `TRUE`; Qatar 3.52, its squared Mahalanobis distance from the mean of the three columns.

10 FAQ

In this section

The question. *The questions this lecture usually leaves behind — answered in two lines each.*

What we do about it. Read these after the home problems; each answer points back to the section where it is explained.

FAQ — Lecture 7

1. Why is $\mathbf{a}^\top \mathbf{S} \mathbf{a}$ a variance and not just a number?

Because it is the variance of the new column $\mathbf{a}^\top \mathbf{x}$ — that is what the formula of Section 1 says, and R confirmed it with `var(X %*% a)`.

2. Can a covariance matrix have a negative eigenvalue?

No: every eigenvalue is the variance of some column ($\lambda_i = \text{Var}(\mathbf{e}_i^\top \mathbf{x})$). A tiny negative value such as -6×10^{-14} is a zero printed with rounding error (Section 4, Home Problem 3).

3. R printed my eigenvector with the opposite sign from the book. Which is right?

Both. $\mathbf{S} \mathbf{e} = \lambda \mathbf{e}$ holds for $-\mathbf{e}$ too. The notes make the biggest entry positive so that printed numbers look the same on every machine; nothing else depends on it (Section 3).

4. Is `eigen(S)$vectors` the matrix $\mathbf{E}$ or $\mathbf{E}^\top$?

$\mathbf{E}$: the eigenvectors are the *columns*. So $\mathbf{E} \mathbf{\Lambda} \mathbf{E}^\top$ is `E %*% diag(lambda) %*% t(E)`, and the new columns are `Xc %*% E` (Sections 3, 4).

5. Why is the first eigenvalue 884.6 bigger than the biggest variance on the diagonal, 801.6?

The income *direction* also includes the part of the other columns that changes together with income. Its share of the total variance (86%) is larger than the income column's share ($801.6/1030 = 78\%$) (Section 4).

6. The textbook's ellipse has half-lengths $c/\sqrt{\lambda_i}$; ours has $c\sqrt{\lambda_i}$. Contradiction?

No. The book writes the ellipse with $\mathbf{A} = \mathbf{S}^{-1}$, whose eigenvalues are $1/\lambda_i$. Put that in and the two formulas are the same (Section 5).

7. Why does Qatar get a larger Mahalanobis distance than Niger when it is closer in the plot?

Qatar moves away from the cloud along the thin direction, where one percentage point counts as $1/3.5$ of a Mahalanobis unit; Niger moves along the long direction, where it counts as only $1/12.2$ (Section 5).

8. Is $\mathbf{S}^{-1/2}$ the only matrix that whitens?

No. Any matrix $\mathbf{W}$ with $\mathbf{W} \mathbf{W}^\top = \mathbf{S}^{-1}$ does, and they all give the same distances; $\mathbf{S}^{-1/2}$ is the symmetric one, built from the eigen-pairs (Section 6).

9. Do I need car to do any of this?

No. `eigen()` and `%*%` do everything printed in this lecture; `car::ellipse()` only draws the picture (Section 5).

Wrap-up

Summary of Lecture 7

- **A weighted total $\mathbf{a}^\top \mathbf{x}$ has variance $\mathbf{a}^\top \mathbf{S} \mathbf{a}$:** read off $\mathbf{S}$, no rows needed. Young+old varies 38.6, young−old 283.9.
- **$\mathbf{S}$ is nonnegative definite** because a variance is never negative; **positive definite** unless one column is a combination of the others (then $|\mathbf{S}| = 0$, no inverse).
- **$\mathbf{S} \mathbf{e}_i = \lambda_i \mathbf{e}_i$, $\|\mathbf{e}_i\| = 1$:** the eigenvectors are the uncorrelated directions, the eigenvalues their variances; $\mathbf{Z} = \mathbf{X}_c \mathbf{E}$ has diagonal covariance. Ours: 884.6, 115.9, 20.1, 9.6; $\mathbf{e}_1$ is mostly income.
- **$\mathbf{S} = \mathbf{E} \mathbf{\Lambda} \mathbf{E}^\top$; $\text{tr}(\mathbf{S}) = \sum \lambda_i = 1030$, $|\mathbf{S}| = \prod \lambda_i$:** one zero eigenvalue makes the determinant zero, and `det()` prints a tiny number that means zero.
- **The Mahalanobis contour is an ellipse** with axes $\mathbf{e}_i$ and half-lengths $c\sqrt{\lambda_i}$; along the long axis a big step is a small Mahalanobis distance. Niger: $d_E = 21.9$, $d_M = 2.2$; Qatar: $d_E = 14.1$, $d_M = 3.7$.
- **$\mathbf{S}^{-1/2} = \mathbf{E} \text{diag}(1/\sqrt{\lambda_i}) \mathbf{E}^\top$. Whitening $\mathbf{X}_c \mathbf{S}^{-1/2}$ gives covariance $\mathbf{I}$,** and the Mahalanobis distance is the straight-line distance afterwards.
- **$\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top \mathbf{S} \mathbf{a} = \lambda_1$ at $\mathbf{e}_1$, $\min = \lambda_p$ at $\mathbf{e}_p$:** stated, checked with 5 000 random directions, not proved.
- **The R to remember:**
 - `eigen()`\$values, `eigen()`\$vectors (columns), and the sign rule;
 - `Xc %*% E` and `E %*% diag(lambda) %*% t(E)`;
 - `zapsmall(cov(Z))`, `mahalanobis()`, `car::ellipse()`.

Next: *Thu, Sep 17 — R Lab 2.*